ĐẠI HỌC HOA SEN

TÔN TRỌNG SỰ KHÁC BIỆT

Introduction to Data Analytics

Faculty of Information Technology

ĐẠI HỌC HOA SEN

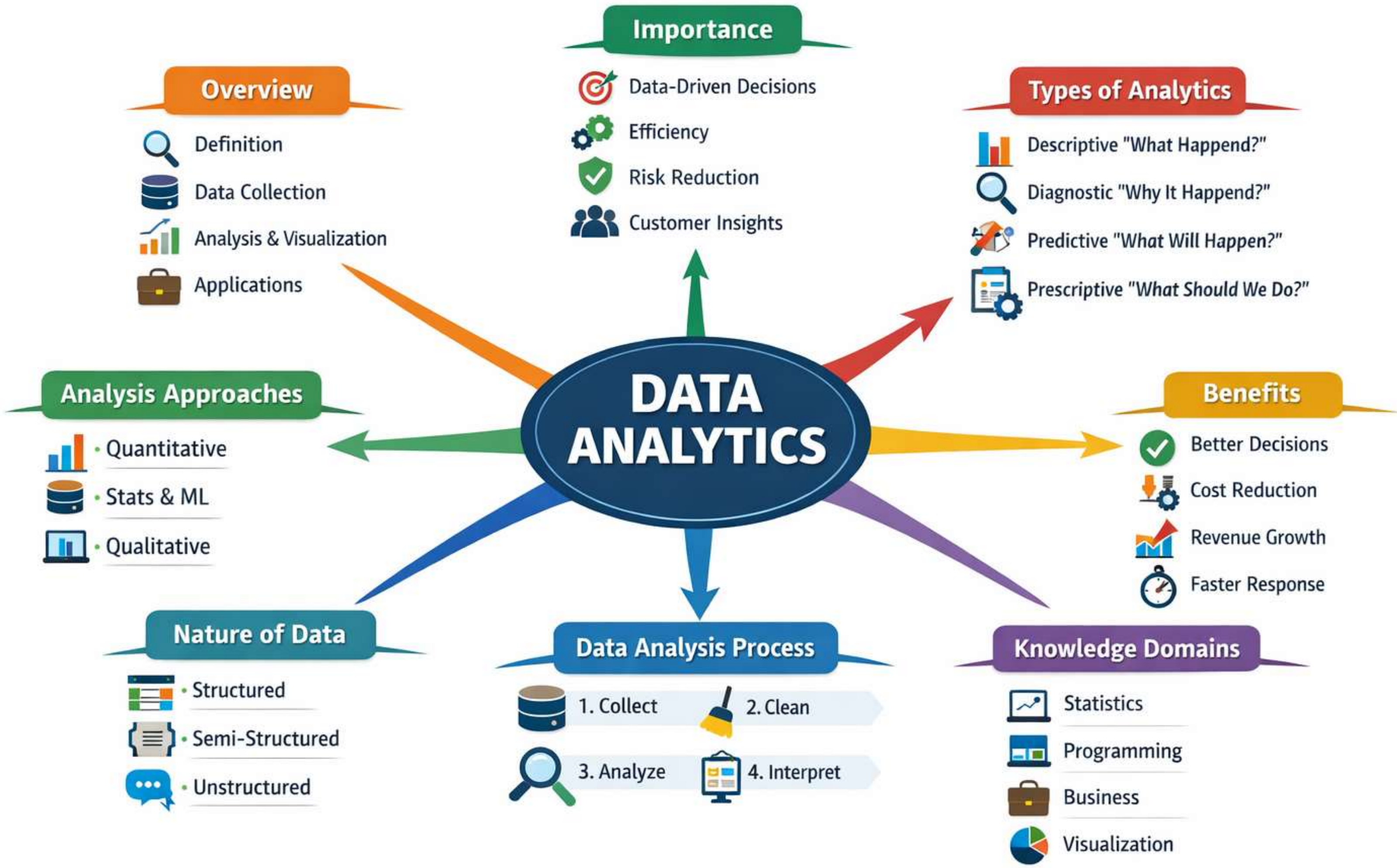
Objectives

- ✓ Data Analytics Overview
- ✓ Types of Data Analytics
- ✓ Benefits of Data Analytics
- ✓ Knowledge Domains
- ✓ Nature of the Data
- ✓ The Data Analysis Process

What is Data Analytics?

- **Data Analytics (DA)** is the systematic process of collecting, cleaning, transforming, analyzing, and interpreting data to extract meaningful insights, support decision-making, and drive business or organizational performance.

- It combines **statistics, mathematics, computer science, and domain knowledge** to turn raw data into actionable information.



DATA ANALYTICS
Overview
Definition
Data Collection
Analysis & Visualization
Applications
Importance
Data-Driven Decisions
Efficiency
Risk Reduction
Customer Insights
Types of Analytics
Descriptive "What Happend?"
Diagnostic "Why It Happend?"
Predictive "What Will Happen?"
Prescriptive "What Should We Do?"
Analysis Approaches
Quantitative
Stats & ML
Qualitative
Benefits
Better Decisions
Cost Reduction
Revenue Growth
Faster Response
Nature of Data
Structured
Semi-Structured
Unstructured
Data Analysis Process
1. Collect
2. Clean
3. Analyze
4. Interpret
Knowledge Domains
Statistics
Programming
Business
Visualization

Data Analytics Overview

- Data Analytics involves:
  - Gathering data from multiple sources (databases, APIs, surveys, sensors, logs)
  - Cleaning and preparing the data
  - Applying analytical techniques (statistical models, algorithms)
  - Visualizing results
  - Interpreting findings for decision-making
- It is widely applied in:
  - Business
  - Healthcare
  - Finance
  - Marketing
  - Education
  - Government
  - Technology
- In modern organizations, Data Analytics supports both **operational decisions (daily actions)** and **strategic decisions (long-term planning)**.

Importance of Data Analytics

- Enables data-driven decision making
  - Improves operational efficiency
  - Identifies trends and patterns
  - Reduces risk and uncertainty
  - Enhances customer experience
  - Creates competitive advantage
  - Supports innovation
- Organizations that leverage analytics effectively often outperform competitors.

Types of Data Analytics

- There are four main types of Data Analytics:
  - **Descriptive Analytics** – What happened?
  - **Diagnostic Analytics** – Why did it happen?
  - **Predictive Analytics** – What will happen?
  - **Prescriptive Analytics** – What should we do?

Descriptive Analytics

- **Question answered:** *What happened?*
  - Descriptive analytics summarizes historical data to understand past performance.
- **Techniques:**
  - Aggregation
  - Data visualization
  - KPI dashboards
  - Reporting
  - Basic statistics (mean, median, variance)
- **Examples:**
  - Monthly sales reports
  - Website traffic dashboards showing number of visitors and page views
  - Revenue by region dashboard
- Descriptive analytics is the foundation of all analytics.

Diagnostic Analytics

- **Question answered:** *Why did it happen?*
  - Diagnostic analytics investigates root causes of events.
- **Techniques:**
  - Drill-down analysis
  - Correlation analysis
  - Data mining
  - Root cause analysis
  - Comparative analysis
- **Examples:**
  - Determining that a drop in sales (the "what") was caused by a specific website outage or a competitor's price drop (the "why")
  - Analyzing why a marketing campaign performed better in one region versus another
- It helps identify relationships and influencing factors.

Predictive Analytics

- **Question answered:** *What will likely happen?*
  - Predictive analytics uses historical data to forecast future outcomes.
- **Techniques:**
  - Regression analysis
  - Time series forecasting
  - Machine learning models
  - Classification algorithms
- **Examples:**
  - Sales forecasting
  - Credit risk prediction
  - Netflix recommending a movie you might like based on your viewing history
  - Forecasting inventory needs for the next quarter
- It estimates probabilities of future events.

Prescriptive Analytics

- **Question answered:** *What should we do?*
  - Prescriptive analytics recommends actions based on predictive insights.
- **Techniques:**
  - Optimization models
  - Simulation
  - Decision analysis
  - Reinforcement learning
- **Examples:**
  - Optimal pricing strategy
  - Supply chain optimization
  - Marketing budget allocation
- It is the most advanced and decision-oriented form of analytics.

Benefits of Data Analytics

- Better decision quality
- Increased profitability
- Faster response to market changes
- Improved customer targeting
- Reduced operational costs
- Enhanced productivity
- Risk mitigation

Knowledge Domains in Data Analytics

- Data Analytics integrates multiple domains:
  - **Statistics and Mathematics** – Probability, hypothesis testing, regression
  - **Computer Science** – Programming (Python, R), databases (SQL), algorithms
  - **Business Knowledge** – Understanding the specific industry (e.g., healthcare, finance, marketing) to interpret what the data actually means in context
  - **Data Management** – Data warehousing, ETL
  - **Visualization** – The ability to communicate findings clearly through charts and graphs
  - **Machine Learning & AI** – Predictive modeling

- A strong data analyst needs both technical and domain expertise.

Nature of the Data

- Data can be classified in several ways:

1. Structured Data
  - Organized in tables (rows & columns)
  - Stored in relational databases
  - Example: sales transactions

2. Semi-Structured Data
  - JSON, XML, logs
  - Flexible schema

3. Unstructured Data
  - Text, images, audio, video
  - Social media posts
  - Emails

Data Types (Statistical Perspective)

- **Quantitative Data**
  - Numerical
  - Discrete (number of customers)
  - Continuous (revenue, temperature)
- **Qualitative Data**
  - Categorical
  - Nominal (gender, region)
  - Ordinal (rating scale)

The Data Analysis Process

Step 1: Problem Definition
- Define objectives
- Identify key questions

Step 2: Data Collection
- Internal systems
- External sources
- Surveys, APIs

Step 3: Data Cleaning
- Handle missing values
- Remove duplicates
- Correct errors

Step 4: Data Exploration (EDA)
- Summary statistics
- Visualization
- Pattern identification

Step 5: Modeling

- Apply statistical or ML techniques

Step 6: Evaluation

- Validate model accuracy
- Performance metrics

Step 7: Interpretation & Communication

- Insights
- Visualization
- Business recommendations

Step 8: Deployment & Monitoring

- Implement solutions
- Track performance

Required Skills

- Statistics & Mathematics
- Python / SQL
- Business understanding
- Communication & storytelling

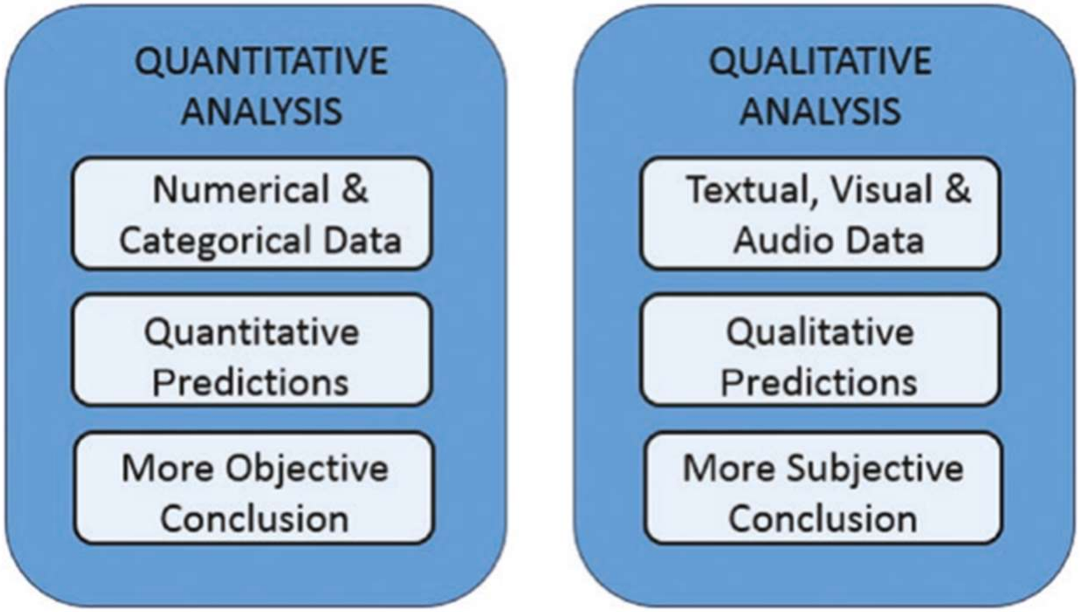
Quantitative and Qualitative Data Analysis

Quantitative Data Analysis

- Focuses on numerical data.
- Methods:
  - Descriptive statistics
  - Inferential statistics
  - Regression
  - Hypothesis testing
  - Machine learning
- Tools:
  - Python
  - R
  - SQL
  - Excel
  - Power BI
  - Tableau
- Best for:
  - Measuring performance
  - Forecasting
  - Statistical validation

Qualitative Data Analysis

- Focuses on non-numerical data (text, interviews, opinions).
- Methods:
  - Thematic analysis
  - Content analysis
  - Sentiment analysis
  - Coding and categorization
- Tools:
  - NVivo
  - Text mining tools
  - NLP libraries
- Best for:
  - Understanding behaviors
  - Exploring motivations
  - Customer feedback analysis



Open Data

- **Kaggle** (www.kaggle.com/datasets)
- **DataHub** (datahub.io/search)
- **Nasa Earth Observations** (https://neo.gsfc.nasa.gov/dataset_index.php/)
- **World Health Organization** (www.who.int/data/collections)
- **World Bank Open Data** (https://data.worldbank.org/)
- **Data.gov** (https://data.gov)
- **European Union Open Data Portal** (https://data.europa.eu/en)
- **Healthdata.gov** (www.healthdata.gov/)
- **Google Trends Datastore** (https://googletrends.github.io/data/)

Summary

- Data Analytics transforms raw data into meaningful insights. It evolves from:
  - Understanding the past (Descriptive)
  - Explaining causes (Diagnostic)
  - Predicting the future (Predictive)
  - Recommending actions (Prescriptive)
- It requires multidisciplinary knowledge and follows a structured analytical process to deliver measurable value to organizations.

Case Study: Retail Sales Decline

- Company experienced 15% sales drop
- Analyze dataset
- Identify root cause
- Recommend actions

- File: DA02_Retail_Analytics_Large_Dataset.csv

Class Activity

- Explore dataset
- Identify trends
- Build simple prediction model
- Present findings

End-to-End Retail Analytics Project

- Apply Descriptive, Diagnostic, Predictive, and Prescriptive Analytics using Python.

PART 1 – Data Understanding & Descriptive Analytics

Step 1: Load Data

```
import pandas as pd
```

```
from google.colab import files
import io
```

```
uploaded = files.upload()
```

```
Choose Files DA02_Reta...Dataset.csv
DA02_Retail_Analytics_Large_Dataset.csv(text/csv) - 8873 bytes, last modified: 2/23/2026 - 100% done
Saving DA02_Retail_Analytics_Large_Dataset.csv to DA02_Retail_Analytics_Large_Dataset.csv
```

```
df = pd.read_csv("DA02_Retail_Analytics_Large_Dataset.csv")
df.head()
```

	Date	Marketing_Spend	Price_Index	Customer_Satisfaction	Competitor_Promo	Sales
0	1/31/2020	27.48	101.83	7.40	0	87.35
1	2/29/2020	24.31	104.54	6.74	0	78.28
2	3/31/2020	28.24	107.02	6.21	0	68.76
3	4/30/2020	32.62	108.46	7.66	0	88.45
4	5/31/2020	23.83	100.20	6.89	0	62.51

Step 2: Summary Statistics

```
df.describe()
```

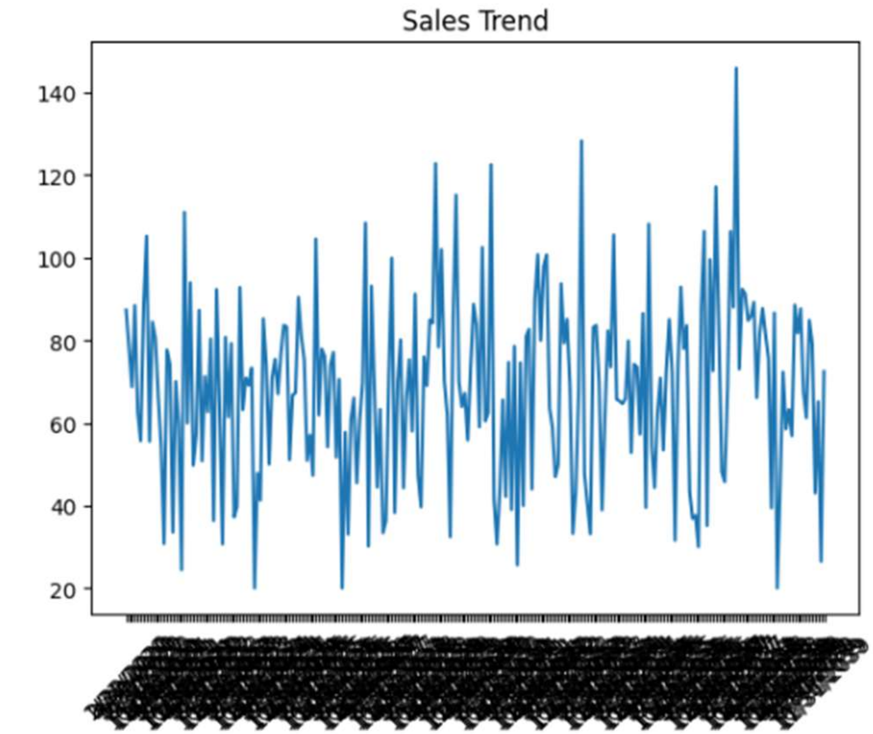
	Marketing_Spend	Price_Index	Customer_Satisfaction	Competitor_Promo	Sales
count	240.000000	240.000000	240.000000	240.000000	240.000000
mean	24.988167	105.142667	7.449083	0.329167	68.243333
std	4.853972	3.959963	0.799913	0.470893	22.199127
min	11.900000	92.030000	5.340000	0.000000	20.000000
25%	21.572500	102.480000	6.890000	0.000000	52.522500
50%	25.310000	105.045000	7.450000	0.000000	69.620000
75%	27.962500	107.732500	7.972500	1.000000	83.200000
max	44.260000	117.320000	9.560000	1.000000	145.930000

Step 3: Sales Trend Visualization

```
import matplotlib.pyplot as plt

plt.figure()
plt.plot(df["Date"], df["Sales"])
plt.xticks(rotation=45)
plt.title("Sales Trend")
plt.show()
```

...



- **Exercise 1**
  - What is average monthly sales?
  - What is the highest sales month?
  - Is there an upward or downward trend?

PART 2 – Diagnostic Analytics

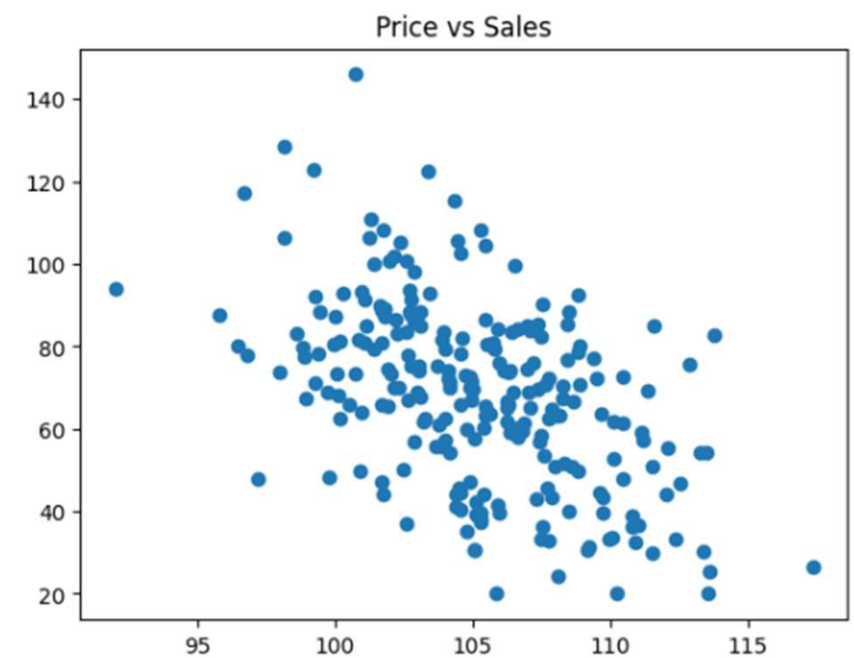
Step 4: Correlation Matrix

```
df.corr(numeric_only=True)
```

	Marketing_Spend	Price_Index	Customer_Satisfaction	Competitor_Promo	Sales
Marketing_Spend	1.000000	-0.083251	0.019126	0.066604	0.657378
Price_Index	-0.083251	1.000000	-0.020106	-0.005813	-0.498884
Customer_Satisfaction	0.019126	-0.020106	1.000000	-0.124494	0.317496
Competitor_Promo	0.066604	-0.005813	-0.124494	1.000000	-0.293754
Sales	0.657378	-0.498884	0.317496	-0.293754	1.000000

Step 5: Relationship Visualization

```
plt.figure()
plt.scatter(df["Price_Index"], df["Sales"])
plt.title("Price vs Sales")
plt.show()
```



Step 6: Competitor Impact

```
df.groupby("Competitor_Promo")["Sales"].mean()
```

Competitor_Promo	Sales
0	72.801739
1	58.953418

dtype: float64

- **Exercise 2**
  - Which variable has strongest impact on sales?
  - Does competitor promotion reduce sales?
  - Is higher satisfaction associated with higher sales?

PART 3 – Predictive Analytics

Step 7: Regression Model

```
#Step 7: Regression Model
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

X = df[["Marketing_Spend", "Price_Index", "Customer_Satisfaction", "Competitor_Promo"]]
y = df["Sales"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)

print("Model Score (R²):", model.score(X_test, y_test))
print("Coefficients:", model.coef_)
```

```
Model Score (R²): 0.9027308793288408
Coefficients: [  2.76421685  -2.464537     7.38118763 -14.32685527]
```

- **Exercise 3**
  - Interpret coefficient of Price_Index.
  - Which variable has largest impact?
  - What does R^2 indicate?

PART 4 – Prescriptive Analytics

Step 8: Scenario Testing

```
#Step 8: Scenario Testing
# Scenario: Increase marketing by 5 units
scenario = model.predict([[30, 105, 8, 0]])
print("Predicted Sales:", scenario)
```

```
Predicted Sales: [91.23006465]
/usr/local/lib/python3.12/dist-packages/sklearn,
  warnings.warn(
```

Step 9: Optimization Simulation

```python
#Step 9: Optimization Simulation
import numpy as np

budgets = np.arange(15, 40, 1)
results = []

for b in budgets:
    pred = model.predict([[b, 105, 8, 0]])
    results.append((b, pred[0]))

best = max(results, key=lambda x: x[1])

print("Optimal Marketing Budget:", best[0])
print("Expected Sales:", best[1])
```

```
Optimal Marketing Budget: 39
Expected Sales: 116.1080162960711
```

- **Exercise 4**
  - What marketing budget maximizes predicted sales?
  - Should the company raise or lower prices?

FINAL LAB DELIVERABLE

- Students must submit:
  - Summary findings
  - Regression interpretation
  - Recommended business strategy
  - Python code (Da02_StuName_StuID.ipynb)


Q&A